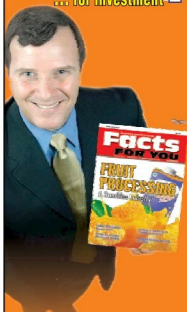


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are universally accepted. This standardisation has given a boost to the use of COTS boards with such peripherals inbuilt. The tedious job of writing device drivers, booting and configuration is already taken care of in such applications.

Automotive electronics. Automotive electronics is a specialised application requiring low power consumption, high efficiency and strong vehicle network communication capabilities.

Rajesh Pathak, senior technology specialist, Delphi Electronics, says, "There are lots of customised microcontrollers available from various vendors which are designed to cater to the needs of the automotive world."

Expressing his thoughts on x86 platforms, Pathak shares, "Though x86 has a very high processing speed (greater than 3 GHz) and a 64-bit variant, it has not found its place in embedded automotive products. For automotive applications, we use a microcontroller with clock speed of 40-256 MHz but it is so specialised for the application that automotive timing needs are not compromised."

For automotive applications, mostly 8-/16-/32-bit reduced instruction set computing (RISC) processors/microcontrollers are used. Pathak says, "There are customised microcontrollers available from various vendors for different electronic control units like engine controller, body controller, safety controller and instrumentation controller." Also, a wide range of development tools are available to allow comfortable product development using these microcontrollers.

Safety-critical applications. You might have experienced the blue-screen-of-death (BSOD) in some signages, kiosks or vending machines. This could be because of power issues or bad software.

Eric Verhulst, CTO, Altreonic, explains, "x86 might work for soft real-time applications, GUIs, etc, where there is ample margin. But for a safety-critical application, it is probably the last choice. It is necessary to keep in mind that when running on a PC, there are tens of dependencies on third-party

software, like drivers. Some drivers, even when proprietary and fully approved, are erroneous."

He shares one experience: "I remember a case where the driver was leaking one byte on every access. In another case, a driver was locking the bus for 10 μ s and it was hard to figure out which driver it was."

Real-time embedded system: does x86 have a place?

Many real-time embedded developers believe that x86 was never meant for embedded applications. It was only meant for desktop and PC-based applications. However, the re-entry of Intel with its Atom clearly hints that Intel never wanted to give away the space. Another thing that embedded developers have strongly realised is that x86 is a good choice for a familiar interface such as Windows operating system.

The fight for performance and power between x86 and ARM has picked up and both are trying to address their weaker areas. The x86 was a preferred architecture for the kind of performance it offered until ARM announced Cortex-A15. The release of Cortex-A15 MPCore is going to have a lot of people re-evaluate their decision based on performance. Some of its astonishing features are 2.5GHz speed, multiple cores, fully cache coherent bus protocol for multisocket systems and support for virtualisation, which is not present even in Intel's Atom processor.

Evaluating the choices for a variety of real-time embedded applications development, ARM leads way over x86. ARM delivers compelling cost-performance and energy-performance ratios. It is a matured architecture licenced by a variety of vendors and the price-competition between them keeps ARM very cost-effective. It has a great support ecosystem of varied users, Open Source tools and numerous vendors. But you never know when Intel pulls out of the bag some new version of Atom! ●

The author was a senior technology journalist at EFY